Regional Quantitative Problem-Solving Circle

Solution Key 6: Stochastic Modeling, Markov Chains & Synthesis (Weeks 11–12)

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Rigorous Proofs & Structural Solutions

Problem 1 (Gambler's Ruin / First-Step Difference Equations — Putnam Caliber)**.** Derive the boundary value recurrence for ruin probability R_k, analyze telescoping differences $(R_{k+1} - R_k)$, and derive exact closed-form expressions for $p = 1/2$ and $p \neq 1/2$.

Proof. Let R_k be the probability of ruin starting from state k. By First-Step Analysis, the process transitions to state $k+1$ with probability p and to state $k-1$ with probability q. The recurrence is:

$$R_k = pR_{k+1} + qR_{k-1}$$

The absorbing boundary conditions are $R_0 = 1$ (immediate ruin) and $R_N = 0$ (immediate victory). Because $p + q = 1$, we can rewrite the left side as $(p+q)R_k$:

$$\begin{aligned} pR_k + qR_k &= pR_{k+1} + qR_{k-1} \\ p(R_{k+1} - R_k) &= q(R_k - R_{k-1}) \\ R_{k+1} - R_k &= \frac{q}{p}(R_k - R_{k-1}) \end{aligned}$$

Define the first-order difference $\Delta_k = R_{k+1} - R_k$. The relation establishes a geometric progression: $\Delta_k = (\frac{q}{p})\Delta_{k-1}$. By recursive substitution down to the base difference $\Delta_0 = R_1 - R_0$:

$$\Delta_i = \left(\frac{q}{p}\right)^i (R_1 - R_0)$$

We isolate R_k by summing the telescoping differences from index 0 to $k-1$:

$$\begin{aligned} \sum_{i=0}^{k-1}(R_{i+1} - R_i) &= R_k - R_0 \\ R_k - 1 &= \sum_{i=0}^{k-1}\left(\frac{q}{p}\right)^i (R_1 - 1) \end{aligned}$$

Case 1 ($p = 1/2 \implies q/p = 1$): The summation reduces to summing the constant 1 exactly k times.

$$R_k - 1 = k(R_1 - 1)$$

We apply the upper boundary $R_N = 0$ to solve for R_1: $0 - 1 = N(R_1 - 1) \implies (R_1 - 1) = -1/N$. Substituting back: $R_k = 1 - \frac{k}{N}$.

Case 2 ($p \neq 1/2 \implies q/p \neq 1$): The summation is a standard finite geometric series:

$$R_k - 1 = (R_1 - 1)\frac{1 - (q/p)^k}{1 - (q/p)}$$

We apply the upper boundary $R_N = 0$ to isolate $(R_1 - 1)$:

$$-1 = (R_1 - 1)\frac{1 - (q/p)^N}{1 - (q/p)} \implies (R_1 - 1) = -\frac{1 - (q/p)}{1 - (q/p)^N}$$

Substituting back into the R_k equation yields the closed-form ruin probability:

$$R_k = 1 - \frac{1 - (q/p)^k}{1 - (q/p)^N} = \frac{(q/p)^k - (q/p)^N}{1 - (q/p)^N}$$

□

Problem 2 (Expected Hitting Times on Symmetric Cyclic Graphs — MIT 6.042 Standard)**.** Derive the recurrence $E_k = 1 + 0.5E_{k-1} + 0.5E_{k+1}$, evaluate $\Delta^2 E_k = -2$, and prove $E_k = k(N - k)$.

Proof. Let E_k be the expected time to reach the absorbing node 0 starting from node k. By First-Step Analysis, taking one step consumes 1 time unit, after which the particle transitions to $k - 1$ or $k + 1$ with equal probability 1/2. Thus:

$$E_k = 1 + \frac{1}{2}E_{k-1} + \frac{1}{2}E_{k+1} \quad \text{for } 1 \leq k \leq N - 1$$

The boundary conditions are $E_0 = 0$ and $E_N = 0$ (since node N is topologically identical to node 0 on the cycle). We rearrange the recurrence to isolate the discrete differences. Multiply by 2:

$$\begin{aligned} 2E_k &= 2 + E_{k-1} + E_{k+1} \\ E_k - E_{k-1} - (E_{k+1} - E_k) &= 2 \end{aligned}$$

Define the first-order forward difference $\Delta E_k = E_{k+1} - E_k$. Substituting this yields:

$$\begin{aligned} -(\Delta E_k - \Delta E_{k-1}) &= 2 \\ \Delta^2 E_k &= -2 \end{aligned}$$

The second-order finite difference is a strictly negative constant (-2). In discrete calculus, a constant second difference implies the solution sequence E_k must be a purely quadratic polynomial of the form:

$$E_k = Ak^2 + Bk + C$$

We apply our boundary conditions to solve for the coefficients. From $E_0 = 0$, we immediately derive $C = 0$. To satisfy $\Delta^2 E_k = -2$, the leading coefficient of the quadratic must be half of the second difference (analogous to the continuous second derivative). Thus, $A = -2/2 = -1$. The equation simplifies to $E_k = -k^2 + Bk$. From the topological wrap-around boundary $E_N = 0$:

$$-N^2 + BN = 0 \implies B = N$$

Substituting B into our polynomial provides the exact expected hitting time:

$$E_k = -k^2 + Nk = k(N - k)$$

□

Problem 3 (Synthesis: Coupling & Stationary Distribution Convergence)**.** Formulate the dual-particle system as a single absorbing Markov chain and derive the exact expected meeting time.

Proof. The total state space of the two particles is strictly vast, but due to the complete symmetry of the complete graph K_M, we can collapse the state space into a binary Markov chain based strictly on spatial intersection. Let the states be S_0 (particles occupy distinct vertices) and S_1 (particles occupy the same vertex). State S_1 is absorbing. We must calculate p, the probability of transitioning from S_0 to S_1 in a single discrete step. Let the particles originate at vertices A and B. There are exactly three mutually exclusive pathways for a collision to occur:

1. Particle A stays at A, and Particle B moves to A. The joint probability is:

$$\frac{1}{2} \times \frac{1}{2(M-1)} = \frac{1}{4(M-1)}$$

2. Particle B stays at B, and Particle A moves to B. Symmetrically, the probability is:

$$\frac{1}{2(M-1)} \times \frac{1}{2} = \frac{1}{4(M-1)}$$

3. Both Particle A and Particle B move simultaneously to a third, identical vertex $v \notin \{A, B\}$. There are exactly $(M-2)$ such available vertices. The joint probability is:

$$(M-2) \times \left(\frac{1}{2(M-1)}\right) \times \left(\frac{1}{2(M-1)}\right) = \frac{M-2}{4(M-1)^2}$$

Since these pathways are mutually exclusive, we sum them to find the total transition probability p:

$$\begin{aligned}
p &= \frac{1}{4(M-1)} + \frac{1}{4(M-1)} + \frac{M-2}{4(M-1)^2} \\
p &= \frac{(M-1)+(M-1)+(M-2)}{4(M-1)^2} \\
p &= \frac{3M-4}{4(M-1)^2}
\end{aligned}$$

Because the system is memoryless and the collision probability p is strictly constant at every step in S_0, the number of steps T required to hit the absorbing state S_1 perfectly follows a geometric distribution. The expected value of a geometric random variable is the reciprocal of its success probability:

$$\mathbb{E}[T] = \frac{1}{p} = \frac{4(M-1)^2}{3M-4}$$

Thus, the expected time for the independent random walks to intersect is strictly bounded. $\square$